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Autonomic Software and Systems Assignment 4

Task 1

1. A rule-based knowledge representation approach can be useful since it works by studying the situation that it is in and deciding based on that information. So, say in an autonomic vehicle an example could be that if the speed limit is 50 then speed up to that, or if there is a stop sign then stop.
2. Learning-based knowledge representation can be useful in self-driving cars since they can't purely rely on the rule-based approach and instead need a way to learn to decide the best course of action. The learning-based approach is based on the evaluation of the usefulness of the systems decisions. When the system makes a decision, it evaluates the usefulness of the decision and learns from it. This is useful since the car would get mixed signals if it worked just based on rules, for example if it got to a speed limit sign and started speeding up but at the same time there is something in front of it, then it needs to know which decision is smarter, going or stopping.
3. Metric optimization with constraints is based on the system having a defined goal and defined limitations. This allows the designer to be able to decide trade-offs for the system so that it can have clear definitions for what it should focus on.
   Machine learning-based techniques are based on learning patterns from data. This means that the model uses the data to build models for predictions, classifications or anything that is based on old data. This means that it can be great at detecting anomalies or predicting coming trends.
   So the main difference between the two methods is mainly that one has set rules for how it should operate while the other learns from old data and uses it to adapt to its own situation.

Task 2

1. Model-based adaptive control is useful in space missions since the spacecraft needs to be able to react in real time to possible changes in the surrounding. The model-

based control is as the name suggests based on models of the system and the environment and gets updated when either is changing.

The supervisory control method is based on a human operating the system. This does not mean that all control is done by the human but that the spacecraft handles the real-time reactions that need to be fast autonomously, while the human contributes by planning the route and such.

The differences between the model-based adaptive control and the supervisory control is that the model-based one is one that reacts purely to the environment and its changes while the supervisory is one that is planned by a human but in case of changes or obstacles that need fast reactions it works autonomously.

2. The Titan probe used autonomy since ground control was impossible due to the 68 minute delay in the signal. So the parachutes used the inertial acceleration sensors to determine when the probe was at critical velocity. It also needed to separate from the heat shield and change from the larger parachutes to the smaller ones. It however was not an adaptive system but instead more of a hard wired one where it used the sensor values to determine a point at where each thing had to happen.
3. It can be used to improve the future missions by making the spacecraft more adaptable to the changing and unknown environment of space. Making spacecrafts capable of guiding themselves and being able to react to changes will make them safer and less likely to fail due to some unforeseen reason. Autonomy could also help with making the spacecraft more fuel efficient by reacting to the environment and by calculating what maneuvers are most fuel efficient.